

Standby for Set-Top Boxes

Set-top boxes have the highest standby power consumption found in home appliances. Thanks to the upcoming change to digital TV formats, many consumers will be forced to either purchase or rent set-top boxes.

Consumer Reports (CR) recently (September 2001) tested a wide range of set-top boxes for satellite systems. For the first time, CR also listed the standby power consumption of the units. We supplemented the data with Lawrence Berkeley National Laboratory measurements of cable boxes and the proposed specifications for Energy Star boxes (see Table 1). Energy Star calls for significant reductions, but not for a few years, when the level will fall to 7 watts for all devices.

Table 1. Standby Consumption

Device	Brand	Watts	kWh/year	Source of Information
Satellite receivers	RCA DS4120RE	10	87	CR*
	EchoStar Dish301	11	96	CR
	Sony SAT-T60	40	348	CR
	Hughes GXCEBOT Philips DSR6000R			
	RCA DWD490RE	35	305	CR
Satellite HDTV models	Hughes E86 Platinum HD	30	261	CR
	Mitsubishi SR-HD500 Toshiba DST-3000			
	RCA DTC-100	35	305	CR
Analog cable boxes	31 measurements	5–25	44–218	www.standby.lbl.gov
Digital cable boxes	29 measurements	9–19	78–165	www.standby.lbl.gov
Energy Star Levels Tier 1 Specifications				
Analog set-top box		<3	<26	Energystar.gov
Satellite TV set-top box		<15	<131	Energystar.gov

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The standby levels range from 10W–40W, but many are above 30W. The highest consumers are units equipped with digital videodisk recorders—often called “TiVo”—designed to record pre-selected program material. These units don’t even have off switches because they are expected to operate continuously, always searching for and downloading material. Cable units are somewhat lower, though all TiVo-like units typically draw from 20W–40W.

Total standby power use in American homes currently is around 60W. This figure includes standby power for televisions, VCRs, microwave ovens, chargers for mobile phones, clock radios, Dust Busters, and other electric devices. However, the widespread introduction of set-top boxes could easily increase standby consumption by 50%. 

—Alan Meier

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